Assignment 2: Detecting Emotions with ML Algorithms

Cross-Dataset Generalization of Facial Expressions

Su Acar (2475901)
Vikram Chandrasekaran (2542544)
Henrique DaSilva Couto (2463334)

5ARE0 – Data Analysis & Learning Methods
Assignment 2 – Group 5
October 25, 2026

Example citations for later on (remove this later): (Goodfellow et al., 2013)
(Goeleven et al., 2008)
(Goeleven et al., 2008; Goodfellow et al., 2013)

1 Introduction and research questions

This report studies facial expression recognition with classical machine learning on two datasets, FER2013 and KDEF, and asks which challenges are specific to each dataset and how well knowledge transfers between them. The FER2013 baseline classifiers are covered in Task 1; Tasks 2 to 6 cover the KDEF study, cross-dataset generalization, the fuzzy classifier, the creative challenge and the real-time demo. *[Replace: One or two sentences summarising the overall main findings once all tasks are done.]*

The primary question is how datasets and feature/model choices affect within-dataset accuracy and cross-dataset generalization. Secondary questions concern which emotions and representations are most transferable, whether interpretable facial geometry is useful, and how reliable a trained model remains under webcam conditions. *[Replace: Add source-supported motivation and APA citations to FER2013/KDEF after Zotero export.]*

2 Datasets, preprocessing and experimental protocol

FER2013. Images are grayscale, 48×48, with seven labels: angry, disgust, fear, happy, sad, surprise and neutral. Use the provided Training split to fit models, PublicTest for model selection/tuning, and untouched PrivateTest *only* for final evaluation.

KDEF. Use only straight/frontal (S) faces, across seven comparable expressions (KDEF "Afraid" maps to FER "fear"). Divide by identity: 80% of subjects for training, 20% for testing, with no subject overlap.

Common representation. Detect and crop KDEF faces with a margin preserving forehead and jaw; convert to grayscale, resize to 48×48 and apply the same normalization as FER. Apply identical feature extraction at training and inference. Fit any scaling, PCA or learned transformation only on training data.

Fuzzy classifier protocol. For Task 4, geometric features were extracted directly from original-resolution frontal KDEF images, rather than the FER-compatible 48×48 images. Both acquisition sessions for each identity remained in the same partition. Model selection used training subjects only, preserving the held-out KDEF subjects for final evaluation.

FER2013 class balance. We use the provided split: 28,709 training images, 3,589 validation images (PublicTest) and 3,589 test images (PrivateTest). The classes are strongly imbalanced (Table 9). Happy is the largest class (7,215 training images) while Disgust has only 436 (1.5% of the training set). Task 1 deliberately uses no class weighting, so that the plain baseline can later be compared with balancing strategies in the creative challenge (Task 5).

[Replace: Detected-face failures, deduplication/quality checks, exact train/validation/test protocol, seeds, tools, versions and potential limitations. Provide one dataset example if it supports a claim.]

3 Task 1: FER2013 baseline (20%)

Features. Three feature types were extracted, each compared with three classifiers ($3 \times 3 = 9$ models):

- **HOG (900 dimensions):** histogram of oriented gradients on the 48×48 image, 9 orientations, 8×8 pixel cells, 2×2 blocks with L2-Hys normalization. HOG encodes local edge orientation (mouth corners, brows, eye shape, wrinkles) and the block normalization gives some robustness to illumination.
- **Facial landmarks (136 dimensions):** the 68 dlib landmarks, centred and scaled to $[-1, 1]$. The whole 48×48 image is passed to dlib as the face box, so the same procedure can be reused in the real-time demo.
- **PCA (100 dimensions):** raw pixels scaled to $[0, 1]$, projected with whitened PCA. PCA is fitted inside the scikit-learn pipeline on the training data only, which avoids leakage into the validation and test sets.

Classifiers. Multinomial Logistic Regression ($C = 1$, standardized inputs), an RBF-kernel SVM ($C = 1$, $\gamma =$ "scale", standardized inputs) and a Random Forest (300 trees, sqrt features, no scaling). All use a fixed random seed (42).

Results on the validation set. Table 1 shows accuracy, balanced accuracy and macro-F1 for all nine combinations. Macro-F1 was the selection criterion, because it weights all seven classes equally and is therefore not dominated by Happy.

Table 1: FER2013 PublicTest (validation) results for all nine feature–classifier combinations, with default settings. The selected model is in bold. Metric heatmaps are in Appendix A.2.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1
HOG	Logistic Regression	0.435	0.397	0.395
HOG	**SVM**	**0.542**	**0.482**	**0.505**
HOG	Random Forest	0.482	0.414	0.440
Landmarks	Logistic Regression	0.470	0.397	0.394
Landmarks	SVM	0.500	0.409	0.401
Landmarks	Random Forest	0.514	0.456	0.474
PCA	Logistic Regression	0.382	0.299	0.287
PCA	SVM	0.477	0.410	0.430
PCA	Random Forest	0.454	0.395	0.423

Best classifier. The RBF-SVM is the best classifier on two of the three feature types (HOG: 0.505 macro-F1; PCA: 0.430) and the best overall. Its non-linear kernel can model the non-linear relation between local gradient patterns and expression, which the linear Logistic Regression cannot: Logistic Regression is the weakest or near-weakest classifier for every feature type (0.287 to 0.395 macro-F1). The Random Forest is competitive only on landmarks, where it is the best classifier (0.474) and clearly beats the SVM (0.401). A plausible reason, which we have not tested, is that the forest can pick out a few informative coordinates from the 136 landmark values, while the RBF kernel, with standardized inputs and equal weight on all coordinates, is diluted by the uninformative ones. The SVM is also by far the most expensive model to train (438 s on HOG, against 36 s for the forest and 6 s for Logistic Regression).

Best features. HOG is the best representation for Logistic Regression and the SVM (for example 0.505 against 0.430 for PCA and 0.401 for landmarks with the SVM). For the Random Forest, landmarks are the best representation (0.474 against 0.440 for HOG). We attribute the HOG result to its capture of local texture and edge information that distinguishes expressions (open mouth, raised or furrowed brows, wrinkles around the eyes and nose), with some invariance to illumination. Landmarks discard texture, and at 48×48 pixels the dlib detector works with very low resolution, so the landmark positions are noisy. Landmarks are also computed with the full image as the face box, which assumes FER crops are well aligned; this is not always true (see the error analysis below). PCA is a holistic linear projection of raw pixels, so it is sensitive to misalignment, pose and lighting.

Hyperparameter tuning and final test result. Each classifier was tuned on the feature set where it did best on validation. The SVM grid covered $C \in \{1, 5, 10\}$ and $\gamma \in \{0.5, 1, 2\} \times 1/900$, tuned on a 10,000-image subsample of the training set; the Logistic Regression grid covered $C \in \{0.01, 0.1, 1, 10\}$; and the Random Forest grid covered maximum depth (unlimited or 25) and minimum leaf size (1, 3 or 5). The best settings were refit on the full training split and scored on validation: the SVM reached 0.562 macro-F1 ($C = 10$, $\gamma = 1/900$), against 0.505 by default; Logistic Regression ($C = 0.01$) reached 0.411; and the landmark Random Forest (maximum depth 25, minimum leaf 1) reached 0.477. Each tuned model was then evaluated once on the private test split (Table 2).

The test score of the SVM is higher than its validation score (0.568 against 0.562 tuned, and 0.505 default). The paired bootstrap difference between the SVM and the landmark Random Forest is +0.081 with interval $[+0.057, +0.105]$, so the SVM's lead on the test split is unlikely to be noise. The gap between accuracy and balanced accuracy on validation (0.542 against 0.482) is caused mainly by the low recall of Disgust (25%).

Which emotions are hardest to distinguish? Averaging per-class F1 over the nine validation models gives

Table 2: Final test performance of each tuned classifier on the PrivateTest split, with 95% bootstrap intervals for macro-F1 (1,000 resamples). The confusion matrices are in Appendix A.

Features	Classifier	Accuracy	Bal. acc.	Macro-F1	95% interval
HOG	SVM ($C = 10$)	0.575	0.550	0.568	[0.544, 0.590]
Landmarks	Random Forest	0.517	0.468	0.487	[0.461, 0.512]
HOG	Logistic Regression	0.438	0.376	0.378	[0.355, 0.400]

the following ordering from hardest to easiest: Fear (0.26), Disgust (0.27), Angry (0.32), Sad (0.38), Neutral (0.43), Surprise (0.62), Happy (0.64). For the SVM on the test set, F1 is 0.75 for Happy and 0.73 for Surprise, and 0.43 to 0.46 for Angry, Sad and Fear.

- Happy and Surprise are relatively easier to classify. Both have distinctive, large-scale configurations: a raised, widened mouth with visible teeth for Happy, and wide-open eyes and mouth for Surprise. Happy additionally has the most training data.
- Fear, Angry and Sad are harder and are confused with each other and with Neutral. The most frequent errors of the validation SVM are Fear to Sad (24% of all Fear), Sad to Neutral (19%), Angry to Sad (18%), Sad to Happy (17%) and Neutral to Sad (17%); the test set shows the same pattern (Appendix A). These classes are expressed through subtle, low-intensity changes around the brows and eyes, which are hard to resolve at 48×48 pixels. Fear also shares features with Surprise (wide eyes, open mouth), and Sad and Neutral faces are visually close, so Neutral is a frequent "sink" class.
- Disgust is rare (436 training images), so the classifier rarely predicts it. On validation the SVM has precision 1.00 but recall 0.25: when it predicts Disgust it is right, but it misses most Disgust faces. On the test set recall is 0.51, but with only 55 test images this estimate is noisy. Class balancing was tested in Task 5 and did not help.

Evidence from error analysis. Illumination and contrast have a small effect on error. We grouped validation images into quintiles of mean brightness and of contrast (pixel standard deviation) and computed the error rate of the SVM in each group. The error rate is about 0.46 to 0.49 for the three darkest quintiles and about 0.42 to 0.44 for the brightest two, and it rises slightly from about 0.44 to 0.48 as contrast increases (Figure 2). The effect is around five percentage points, so illumination matters but is not the main source of error.

Figure 3 shows recurring problems that are properties of the dataset rather than of the model: *ambiguous or questionable labels* (a screaming infant labelled Angry and predicted Happy; several wide-eyed, open-mouthed faces labelled Neutral or Fear), *occlusion and context* (hands over the face, hats, two faces in the crop), *non-frontal pose and head tilt*, which break the aligned-face assumption of all three feature types, and *population and artefacts* (infants and elderly faces, overlaid text or watermarks). The gallery is a random sample of errors, so it is qualitative evidence, not a count of mislabelled images.

Duplicates. FER2013 contains exact copies of images across splits, and some copies carry different labels, which is direct evidence of label noise. Of the 3,301 non-duplicated test images, the SVM reaches 54.2% accuracy and 0.505 macro-F1; the 288 duplicated test images (8.0%) reach 95.5% accuracy and 0.943 macro-F1, and 96.9% of the duplicates carry the same label as the training copy. The reported test score of 0.568 macro-F1 is therefore optimistic, and the non-duplicated score of 0.505 is the fairer estimate. On validation, 280 of 3,589 images (7.8%) also appear in training, and removing them lowers the SVM's macro-F1 from 0.505 to 0.438. The drop is largest for the Random Forests (0.097 to 0.110), which memorize training samples, and smallest for Logistic Regression and landmark SVM (0.008 to 0.010). We treat cleaning the training set as a data curation step in Task 5.

Summary and limitations. The tuned HOG + RBF-SVM is the best FER2013 baseline (test: 0.575 accuracy, 0.550 balanced accuracy, 0.568 macro-F1; 0.505 macro-F1 without duplicates). HOG is the strongest feature type for Logistic Regression and the SVM, and landmarks are the strongest for the Random Forest; the SVM is the strongest classifier on HOG and PCA. Happy and Surprise are recognized reliably, while Fear, Angry, Sad and Disgust are not. For the demo, prediction time is 0.15 ms per image for Logistic Regression, 32 ms for the SVM, and 27 ms plus 11 ms of feature extraction for the Random Forest. Limitations: (i) the SVM was tuned on a 10,000-image subsample; (ii) the final test was run once per model, with no further selection; (iii) no class balancing was used, which disadvantages Disgust; (iv) the landmark features depend on a detector applied to 48×48 images with the whole image as the face box.

4 Task 2: KDEF dataset study (15%)

Setup. We retrained the three classifiers on HOG features, the best FER2013 representation (Task 1), using the 980 frontal KDEF images of 70 subjects. Images were preprocessed with the provided script: face detection, a

20% margin around the detected box, grayscale conversion and resizing to 48×48. The split is 80/20 by subject identity: 56 subjects (784 images) for training and 14 subjects (196 images) for testing, with 28 test images per emotion. No subject appears in both sets.

Model selection. Hyperparameters were selected by 5-fold cross-validation on the 56 training subjects, grouped by subject so that no person appears in both folds of a split, scored by macro-F1. The grid covered the feature scaler, the regularization constant C (Logistic Regression and SVM), γ (SVM), and tree depth, leaf size and number of trees (Random Forest). Each selected configuration was refit on all 784 training images and evaluated once on the 196 test images. The selected settings were: Logistic Regression with a power-transform scaler and $C = 0.01$; RBF-SVM with a power-transform scaler, $C = 10$ and $\gamma = 10^{-4}$; and Random Forest with maximum depth 10.

Table 3: KDEF results with the HOG representation. CV: subject-grouped 5-fold cross-validation on the 56 training subjects. Test: 14 held-out subjects (196 images).

Classifier	CV macro-F1	Test accuracy	Test bal. acc.	Test macro-F1
Logistic Regression	0.669	0.577	0.577	0.581
SVM	**0.679**	0.602	0.602	0.604
Random Forest	0.611	**0.622**	**0.622**	**0.614**

Why KDEF is easier or harder than FER2013. On the same HOG features and classifiers, KDEF scores are higher: the best cross-validated macro-F1 is 0.679 on KDEF against 0.505 on FER2013 validation. The test gap is much smaller (0.604 against 0.568), and with 196 test images a 95% interval on accuracy is about ± 0.07, so the test comparison is only indicative. The likely reasons are that KDEF has controlled studio lighting, a frontal pose, exaggerated posed expressions and balanced classes (28 test images per emotion). FER2013 is in-the-wild, unevenly labelled and strongly imbalanced (Section 2). KDEF is also smaller (70 people, one pose), so it does not test robustness to pose or lighting.

Which models change ranking. On FER2013 validation the order is SVM (0.505), Random Forest (0.440), Logistic Regression (0.395). On KDEF the order is SVM (0.679), Logistic Regression (0.669), Random Forest (0.611) in cross-validation, and Random Forest (0.614), SVM (0.604), Logistic Regression (0.581) on the test subjects. The Random Forest moves from second on FER2013 to last in cross-validation and first on the test subjects, so its position depends on the evaluation set. Given the test size, the test differences of 0.01 to 0.03 are not reliable. The SVM is the most consistent winner in cross-validation and on FER2013.

Easiest and hardest emotions. For the SVM on the test subjects, F1 is highest for Happy (0.93) and lowest for Fear ("afraid", 0.33), with Neutral (0.66), Disgust (0.65), Surprise (0.62), Sad (0.55) and Angry (0.50) in between. Happy and Surprise are the emotions with the clearest mouth and eye shapes, which is consistent with FER2013 (Section 3). The KDEF confusion matrices are in Appendix A. Which emotions transfer between the two datasets is discussed in Section 5.

5 Task 3: Cross-dataset generalization (20%)

Adaptations. The best Task 2 configuration (HOG features, power-transform scaler, RBF-SVM with $C = 10$ and $\gamma = 10^{-4}$) was retrained on all 980 frontal KDEF images after conversion to the FER format: face crop with a 20% margin, grayscale, 48×48, scaled to $[0, 1]$. FER images were only scaled to $[0, 1]$. The same HOG extractor was used for both datasets, and the seven KDEF labels were mapped one-to-one onto the FER labels ("afraid" to "fear"). Nothing was fitted on FER, including the scaler. The FER→KDEF direction in the protocol has not been run yet (see Table 4).

Table 4: Within- and cross-dataset performance. The KDEF→KDEF row is the Task 2 grouped cross-validation; the KDEF→FER row covers all 35,887 FER2013 images, not only PrivateTest.

Train → evaluation	Accuracy	Bal. acc.	Macro-F1
FER → FER (SVM, PrivateTest)	0.575	0.550	0.568
KDEF → KDEF (grouped CV)	0.686	0.686	0.679
KDEF → FER (all FER2013)	0.225	0.215	0.195
FER → KDEF	*[Replace: not yet run]*		

Domain shift. Domain shift means that the training and test distributions differ, $P_{\text{KDEF}}(x, y) \neq P_{\text{FER}}(x, y)$. Three components are present. Covariate shift in $P(x)$: studio photographs with controlled lighting and framing, versus web-sourced images with varying lighting, pose, occlusion and crops. Label shift in $P(y)$: KDEF is

balanced (14.3% per emotion), while FER2013 has 25.0% Happy and 1.5% Disgust. Concept shift in $P(y \mid x)$: KDEF expressions are posed and exaggerated, whereas many FER2013 expressions are subtle and some labels are debatable (Section 3).

Why performance degrades. Accuracy falls from 0.602 on the KDEF test subjects (0.686 in cross-validation) to 0.225 on FER2013. Three observations explain part of the drop. First, the predicted class mix does not match FER2013: Neutral has a recall of 0.012 and Disgust a precision of 0.020. Second, the images differ in statistics: mean brightness is 0.507 in FER2013 against 0.327 in KDEF, and mean contrast (pixel standard deviation) is 0.212 against 0.161. A logistic regression on the HOG features identifies the source dataset with 98.8% cross-validated accuracy, so the representation carries a strong dataset signature. Third, 980 studio images of 70 people cannot cover the variety of FER2013, and HOG compares fixed image regions, so differences in face position and scale change the features directly.

Which emotions transfer. Table 5 compares the per-emotion F1 of the KDEF-trained model on KDEF (grouped cross-validation) and on FER2013. Happy and Surprise keep the most performance. Disgust and Neutral collapse, which is partly caused by their rarity in FER2013 (Table 9).

Table 5: Per-emotion F1 of the KDEF-trained SVM: in-domain (KDEF, grouped CV) and cross-domain (FER2013).

Emotion	F1 on KDEF	F1 on FER2013
Happy	0.89	0.38
Surprise	0.71	0.31
Sad	0.50	0.24
Angry	0.66	0.19
Fear	0.46	0.18
Disgust	0.69	0.04
Neutral	0.66	0.02

Which HOG features transfer. For each HOG feature we correlated its average value across the seven emotions in KDEF with the same pattern in FER2013. The best transferring features (correlations 0.92 to 0.95) lie in the central part of the face grid. The worst (correlations -0.75 to -0.88) lie at the edges and corners of the grid, such as the top-right and bottom-right cells. Only HOG was evaluated across datasets, so we cannot say whether landmarks or PCA transfer better. The result is also based on a single random seed and a single KDEF pose.

6 Task 4: Explainable fuzzy classifier (10%)

Features and protocol. We extracted seven geometric features from the 68 dlib facial landmarks: mouth width, mouth opening, left and right eye opening, eyebrow height, mouth width-to-height ratio (*smile ratio*), and mouth-corner asymmetry. Most distances were normalized by face width, while eye opening was normalized by eye width. All 980 frontal KDEF images were processed successfully, without landmark detection failures.

The predefined subject-independent split contained 56 training subjects (784 images) and 14 held-out test subjects (196 images). Within the training partition, 44 subjects (616 images) were used for fitting and 12 (168 images) for validation. MinMax scaling was fitted only on the corresponding training partition.

Fuzzy model. We trained seven one-versus-rest Takagi–Sugeno fuzzy models with PyFUME, one for each emotion. Each model uses Gaussian membership functions in its rule antecedents and a learned linear consequent. The numerical outputs are combined through fuzzy inference, and the emotion with the highest output score is selected. These scores are not calibrated probabilities.

We compared two and three clusters per emotion using macro-F1 on validation subjects. The three-cluster configuration was selected before testing and subsequently refitted on all 784 training images.

Table 6: Subject-independent KDEF fuzzy classification results.

Split	Clusters	Images	Acc. (%)	Bal. acc. (%)	Macro-F1
Validation	2	168	71.43	71.43	0.6881
Validation	3	168	73.21	73.21	0.7087
Final test	3	196	63.27	63.27	0.6082

Rule interpretation. The final system contains 21 fuzzy rules, three per emotion. A representative rule from the happiness model is:

> *Happy, Rule 1 (abridged):* IF mouth width belongs to cluster 1 (Gaussian centre 0.762) AND mouth opening belongs to cluster 1 (centre 0.204), together with the remaining geometric conditions, THEN calculate the happiness score using the fitted linear consequent.

This rule dominated 94.6% of happy training images but only 2.2% of other images. Its relatively wide-mouth configuration is consistent with smiling. The strongest rules for fear and sadness showed weaker separation, and overlapping membership functions limit their linguistic interpretation.

Evaluation and limitations. On unseen subjects, happiness (28/28), surprise (26/28) and disgust (23/28) achieved the highest recall. Fear (7/28) and sadness (8/28) were more difficult: 14 fearful faces were classified as surprise, and 12 sad faces as neutral. The roughly ten-percentage-point accuracy decrease from validation to testing indicates limited generalization across individuals. Although the fuzzy antecedents are inspectable, the linear consequents and interactions between the seven models complicate explanations of individual predictions.

7 Task 5: Creative challenge (25%)

We started from the Task 1 baseline and tested one change at a time, using a fast Logistic Regression and the validation split (PublicTest) to decide. The test split was used once, for the final model.

Changes tested.

- **Feature engineering (kept).** The 136 landmark values were appended to the 900 HOG features (1,036 features). HOG describes texture around the mouth and eyes, and landmarks describe the shape of the face, so the two carry different information.
- **Data cleaning (kept).** Training images were fingerprinted with MD5 hashes. Duplicates with the same label were reduced to one copy, and images whose copies disagreed in label were removed (107 images), reducing the training set from 28,709 to 27,434 images. The gain is within the noise level (Section 3).
- **Class balancing (rejected).** Balanced class weights lowered both macro-F1 and Disgust F1, so the switch was left off.
- **Augmentation (not run).** Horizontal-flip copies were considered but not run. They would double the training set, and the RBF-SVM training time is already the limiting factor.

Table 7: Validation (PublicTest) ablation with Logistic Regression. Each row adds one change to the rows above it.

Variant	Accuracy	Macro-F1	Disgust F1	Training images
Baseline (HOG)	0.435	0.395	0.324	28,709
+ landmarks	0.497	0.453	0.325	28,709
+ cleaning (kept)	0.500	0.455	0.325	27,434
+ class balancing (rejected)	0.469	0.424	0.242	27,434

Final model. The tuned Task 1 SVM ($C = 10$) was retrained on the kept changes: 1,036 features and the cleaned training set. Because the feature count rose from 900 to 1,036, γ was rescaled by 900/1036 to keep the kernel width comparable ($\gamma \approx 9.7 \times 10^{-4}$). This rescaling is a heuristic and was not tuned. On validation, the model reaches 0.607 accuracy and 0.598 macro-F1, against 0.562 for the tuned Task 1 SVM. On the test split, evaluated once, it reaches 0.594 accuracy, 0.572 balanced accuracy and 0.592 macro-F1, against 0.568 for the Task 1 SVM.

Interpretation. Adding landmarks is the main gain (+0.058 macro-F1 for Logistic Regression), and it is consistent with the landmark-based Random Forest in Task 1, which is the best classifier on landmarks. Cleaning changes the score by 0.002, which we treat as no effect. Class balancing hurt, even though Disgust recall was low (0.25) in Task 1. The notebook does not analyze why, so we leave the mechanism open. The test gain of 0.024 macro-F1 over Task 1 is not tested for significance, and the Task 1 bootstrap interval for macro-F1 is about ±0.023. Two limitations apply. The ablation uses Logistic Regression, not the final SVM, and each step is a single run. The test split still contains the 8% of images that also appear in training (Section 3). KDEF-derived features were not used in the final model.

8 Task 6: Real-time webcam demo (10%)

Adapt the best selected model into a command-line executable Python file that opens the laptop webcam and performs live emotion recognition. Preserve the exact training face crop, alignment, feature extraction, normalization and class mapping in the inference path.

[Replace: State demo command, chosen model/.pkl, hardware/OS, measured or observed FPS/latency (if available), stability, real-world behavior, lighting/pose/occlusion issues and what was fine-tuned. Explain whether the model performed well enough, and why.]

9 Discussion, limitations and conclusion

[Replace: Synthesize: best features/classifiers on each dataset; relative difficulty of each emotion; changes of ranking; strongest and weakest transfers; fuzzy interpretability versus performance; benefits and costs of creative improvements; webcam readiness. Contrast dataset-specific and common limitations. Draw conclusions only from actual measurements and concrete error examples.]

Declaration of AI tool use

During the preparation of this work, I/We used [NAME TOOL / SERVICE / VERSION OF AI TOOL] in order to [REASON]. The following parts of the assignment were affected/generated by AI tool usage: [INTRODUCTION / METHODS / xxx, DISCUSSION]. After using this tool/service, [NAME STUDENT(S)] evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [NAME STUDENT(S)] take(s) full responsibility for the content of their work.

References

Goeleven, E., De Raedt, R., Leyman, L., & Verschuere, B. (2008). The Karolinska Directed Emotional Faces: A validation study. *Cognition & Emotion*, *22*(6), 1094–1118. https://doi.org/10.1080/02699930701626582

Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., Zhou, Y., Ramaiah, C., Feng, F., Li, R., Wang, X., Athanasakis, D., Shawe-Taylor, J., Milakov, M., Park, J., … Bengio, Y. (2013). Challenges in Representation Learning: A Report on Three Machine Learning Contests. In M. Lee, A. Hirose, Z.-G. Hou, R. M. Kil, D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi, & G. Weikum (Eds.), *Neural Information Processing* (pp. 117–124, Vol. 8228). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-42051-1_16

A Confusion matrices for reported models

Confusion matrices for every model reported in the main text, exported by the notebooks. FER→KDEF has not been run yet: *[Replace: add its confusion matrix once run]*.

A.1 Dataset and error-analysis figures (Tasks 1 and 3)

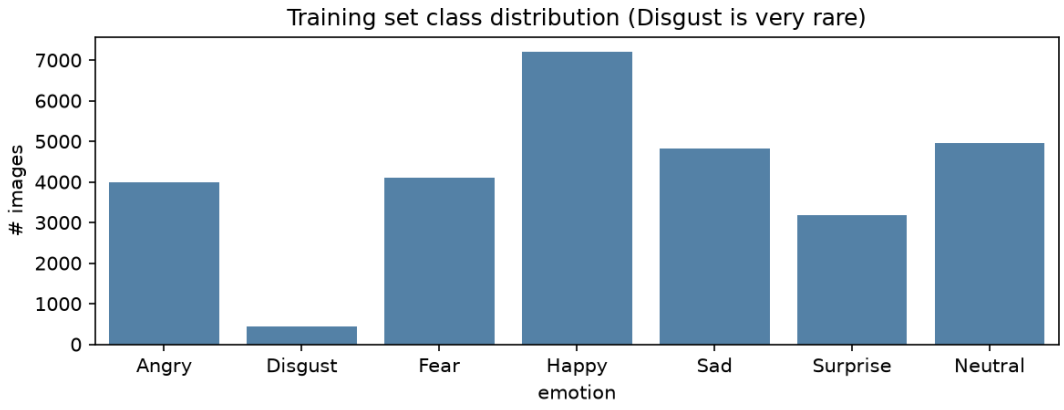


Figure 1: FER2013 training set: class distribution.

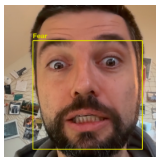
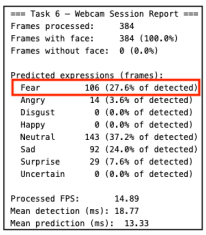
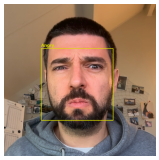
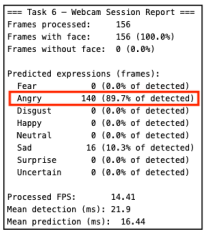
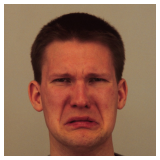
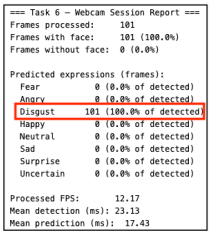
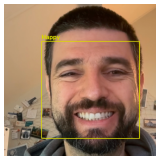
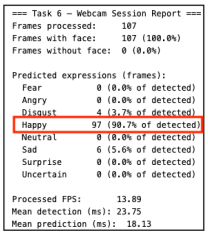
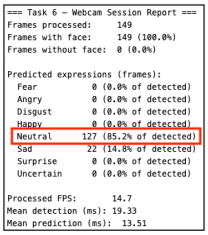
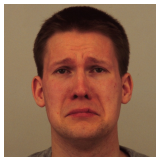
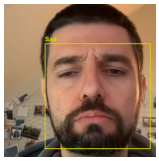
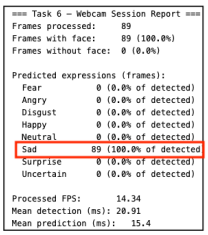
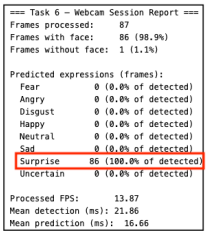
Posed expression	KDEF reference	Webcam capture	Prediction
Fear			
Angry			
Disgust			
Happy			
Neutral			
Sad			
Surprise			

Table 8: Qualitative comparison of posed KDEF expressions, corresponding webcam expressions, and fuzzy classifier predictions. KDEF subject M11, acquisition A, frontal pose. This illustrative experiment does not establish classification accuracy.

Table 9: FER2013 class counts per split.

Split	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Train	3,995	436	4,097	7,215	4,830	3,171	4,965
Validation	467	56	496	895	653	415	607

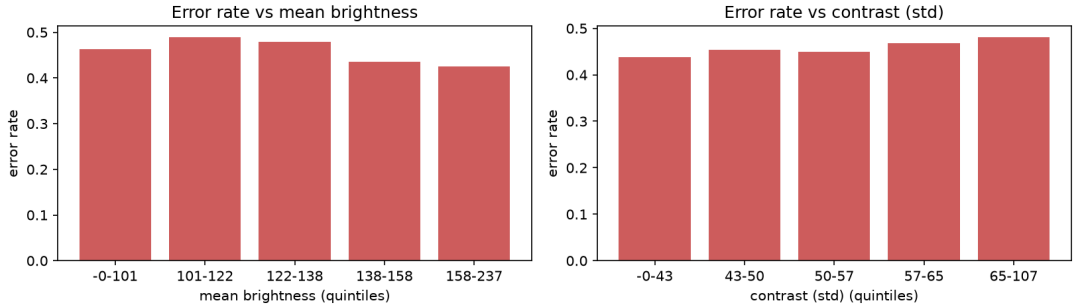


Figure 2: Error rate against lighting (HOG + SVM, validation split).

A.2 Validation metric heatmaps (FER2013)

A.3 FER2013 baseline models (PublicTest; tuned models on PrivateTest)

A.4 KDEF study, cross-dataset and creative models

A.5 Fuzzy classifier: held-out KDEF subjects

Table 10: Final fuzzy classifier confusion matrix (counts; 196 test images from 14 unseen subjects). Rows are true labels and columns are predictions.

True / Pred.	Af	An	Di	Ha	Ne	Sa	Su
Afraid	7	1	2	0	2	2	14
Angry	0	14	10	0	2	2	0
Disgust	0	3	23	1	0	1	0
Happy	0	0	0	28	0	0	0
Neutral	0	3	0	0	18	6	1
Sad	0	6	2	0	12	8	0
Surprise	0	0	0	0	1	1	26

Af: afraid; An: angry; Di: disgust; Ha: happy; Ne: neutral; Sa: sad; Su: surprise.

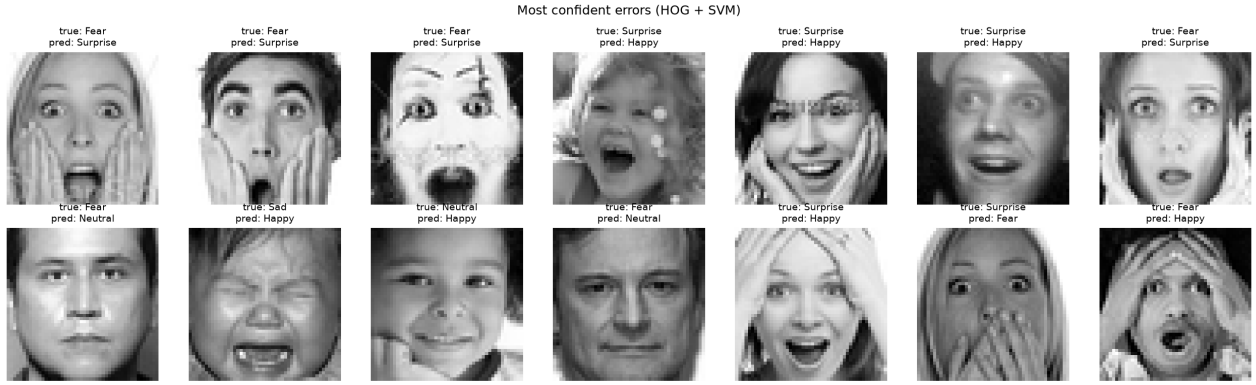


Figure 3: Random sample of misclassified FER2013 validation images.

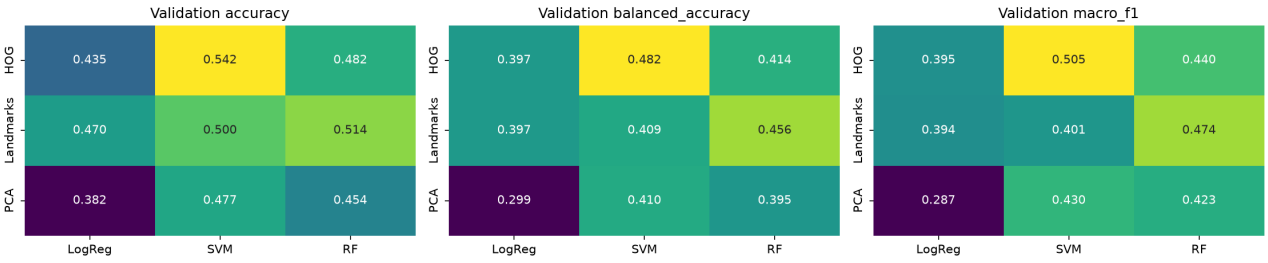


Figure 4: Metric heatmaps of each feature–classifier combination on the FER2013 PublicTest (validation) split.

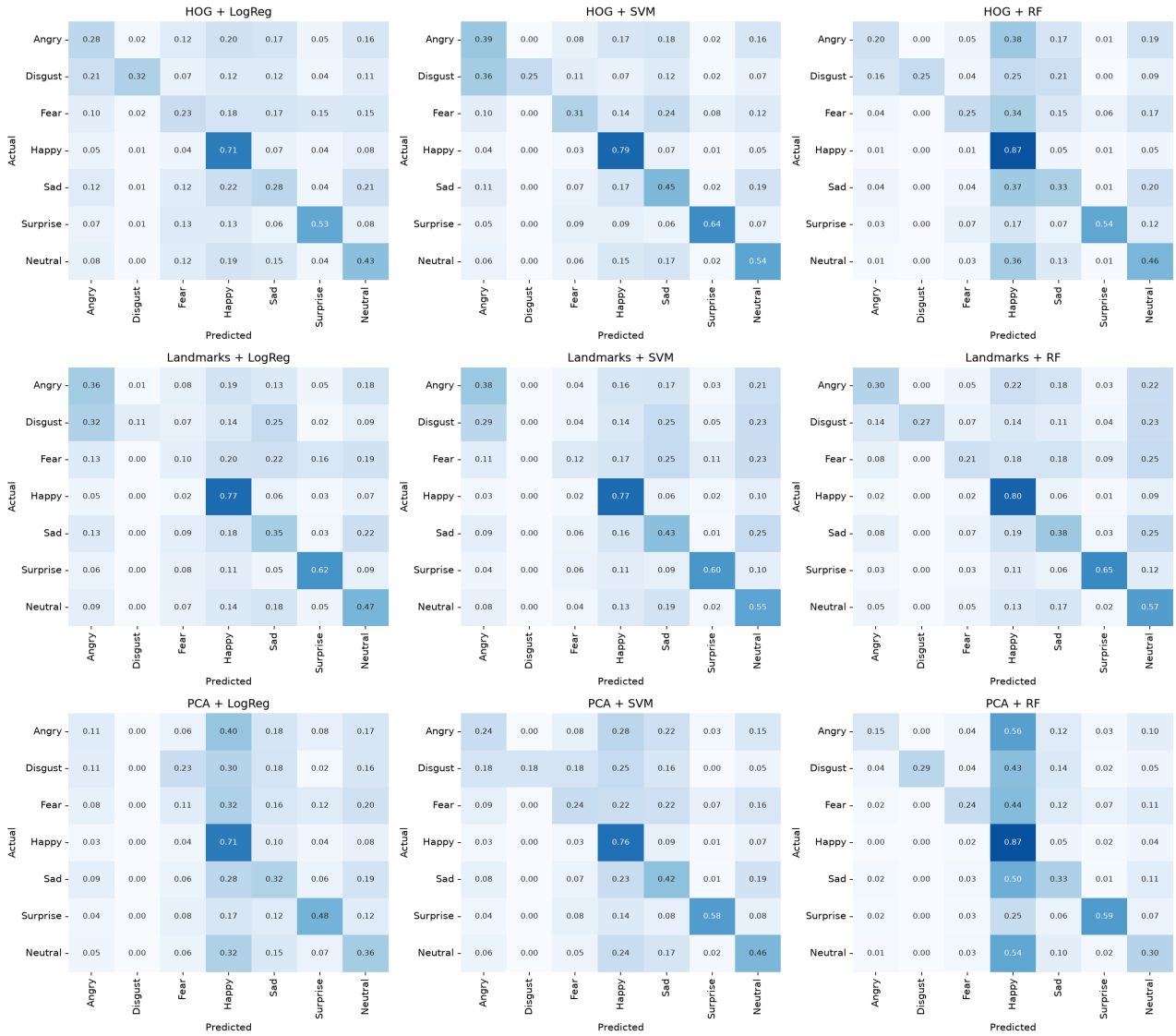


Figure 5: Confusion matrices of all nine FER2013 baseline models on the PublicTest (validation) split. Rows are true labels and columns are predictions; values are row-normalized.

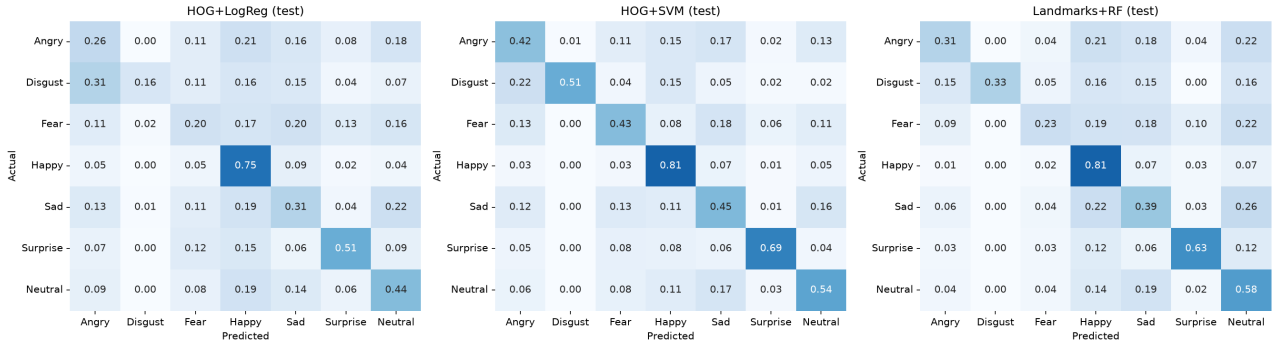


Figure 6: Confusion matrices of the three selected FER2013 models on the PrivateTest split, used for the final evaluation (Table 2). Rows are true labels and columns are predictions; values are row-normalized.

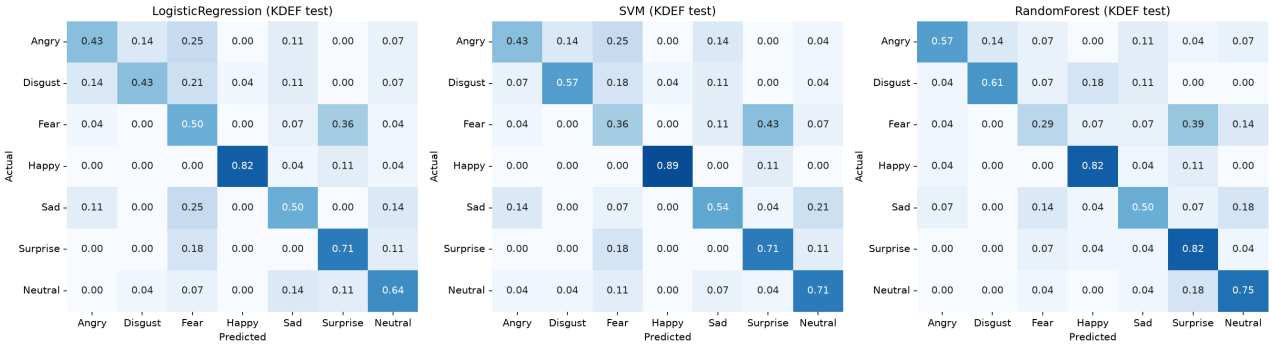


Figure 7: KDEF test subjects (196 images): Logistic Regression, SVM and Random Forest with HOG features, left to right.

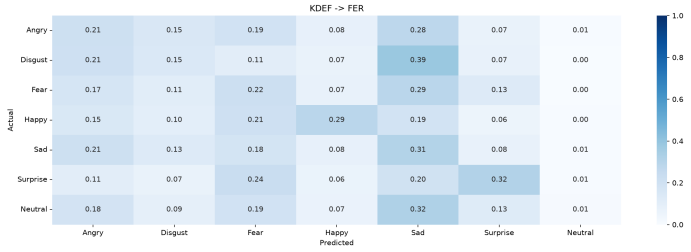


Figure 8: KDEF→FER: the Task 2 SVM trained on all KDEF frontal images, evaluated on all FER2013 images. Rows are true labels, row-normalized.

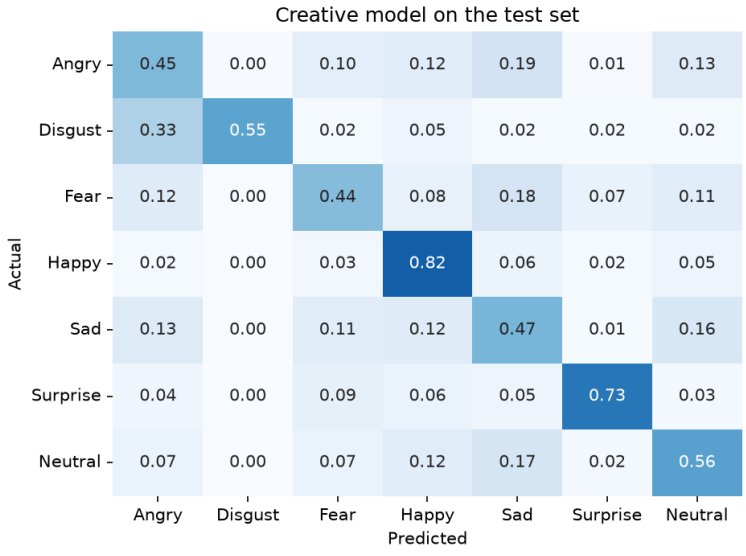


Figure 9: Creative-challenge final model (SVM on HOG + landmarks, cleaned training set), PrivateTest split.